Combining Retrieval-Augmented Text Generation with LLMs for Reading Content Recommendations

WYDZIAŁ ELEKTRONIKI I TECHNIK INFORMACYJNYCH

Sooyeon Kim and Piotr S. Maciąg

Institute of Computer Science, Warsaw University of Technology, Poland

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Motivation

Large Language Models (LLMs) can generate educational reading content, but they often lack factual grounding and precise control over reading difficulty.

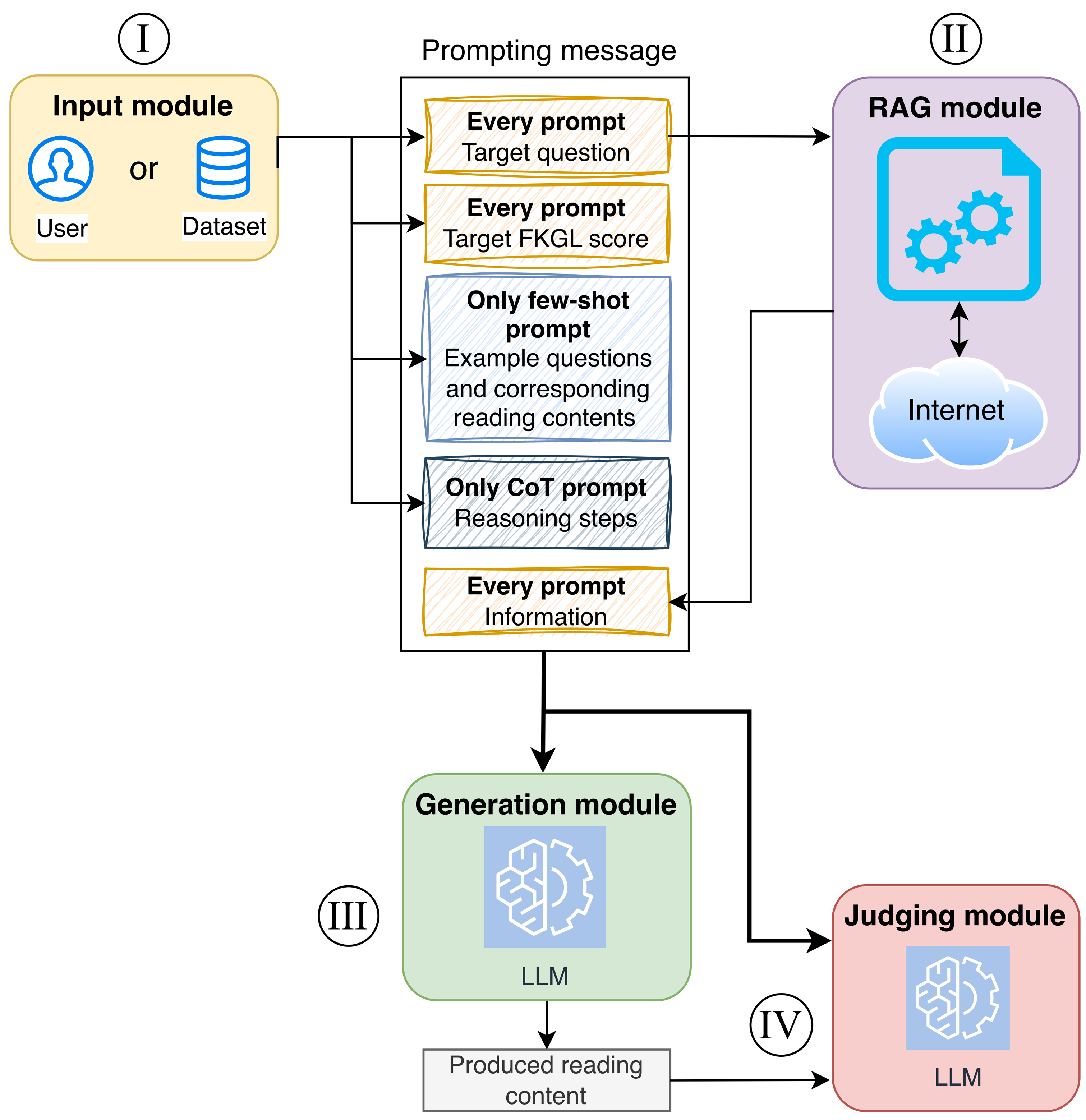
This work proposes a system for **personalized reading content generation** that combines LLMs with **Retrieval-Augmented Generation** and targets a user-specified readability level.

1. Users provide a **question** and a desired **Flesch–Kincaid Grade Level (FKGL)**.
2. RAG retrieves from the Internet external evidence relevant to the question.
3. A selected LLM generates reading material tailored to the requested complexity.
4. Another LLM-as-a-Judge model evaluates the answer generated in step 3.

Main Contributions

- A modular architecture for personalized reading recommendations using **Input**, **RAG**, **Generation**, and **Judging** components.
- Comparison of **zero-shot**, **few-shot**, and **Chain-of-Thought** prompting.
- Text generated using one of LLMs:**Meta LLaMA 4 Scout**, **LLaMA 3.1–8B–Instant**, and **Google Gemma2–9B**.
- Automatic evaluation using **LLM-as-a-Judge**, relevance, groundedness, and FKGL tolerance ranges.

System Architecture and Methodology



Module	Role in the proposed architecture
Input	Receives the user question, the target FKGL score, and optional prompt elements. Tested prompting techniques were **Zero-shot**, **Few-shot**, **Chain-of-Thought**.
RAG	Retrieves external information from the Internet for the target question, including factual snippets, links, or relevant web-page content.
Generation	Uses the selected LLM and the constructed prompt to produce personalized reading content.
Judging	Evaluates the generated reading content with GPT-4.1 used as an LLM-as-a-Judge.

The Generation module was tested with three LLMs: **Meta LLaMA 4 Scout**, **LLaMA 3.1–8B–Instant**, and **Google Gemma2–9B**. The models were hosted through GroqCloud. The RAG module used Tavily for Internet search, while the Judging module used **GPT-4.1**.

Dataset and Evaluation

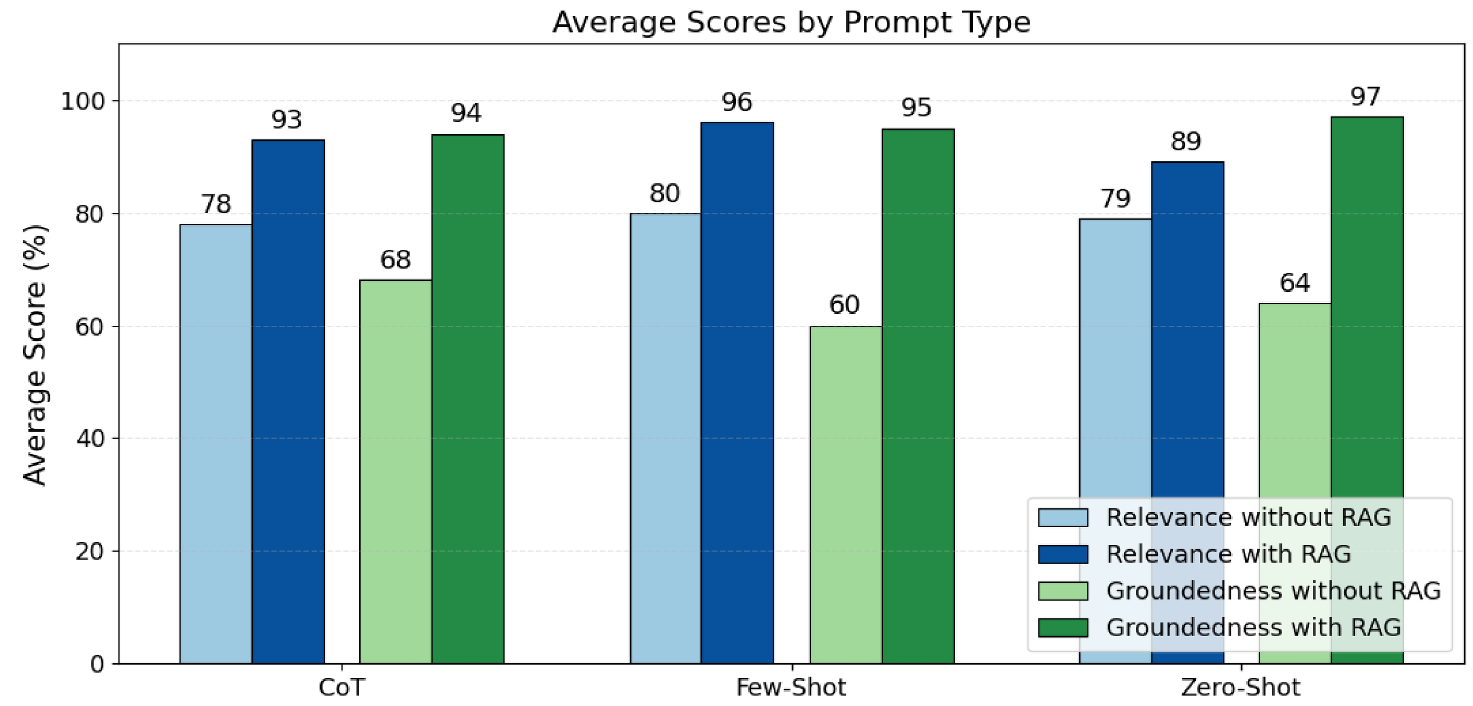
Experiments used a subset of **30 questions** from the *Natural Questions* dataset. Each question was assigned a target FKGL score from **2** to **12**. The quality of answers was measured using:

- **Relevance:** whether the answer directly addresses the question.
- **Groundedness:** whether the answer is supported by retrieved facts.
- **FKGL alignment:** whether generated content matches the requested complexity.

$$\mathrm{FKGL} = 0.39 \times \frac{\text{words}}{\text{sentences}} + 11.8 \times \frac{\text{syllables}}{\text{words}} - 15.59$$

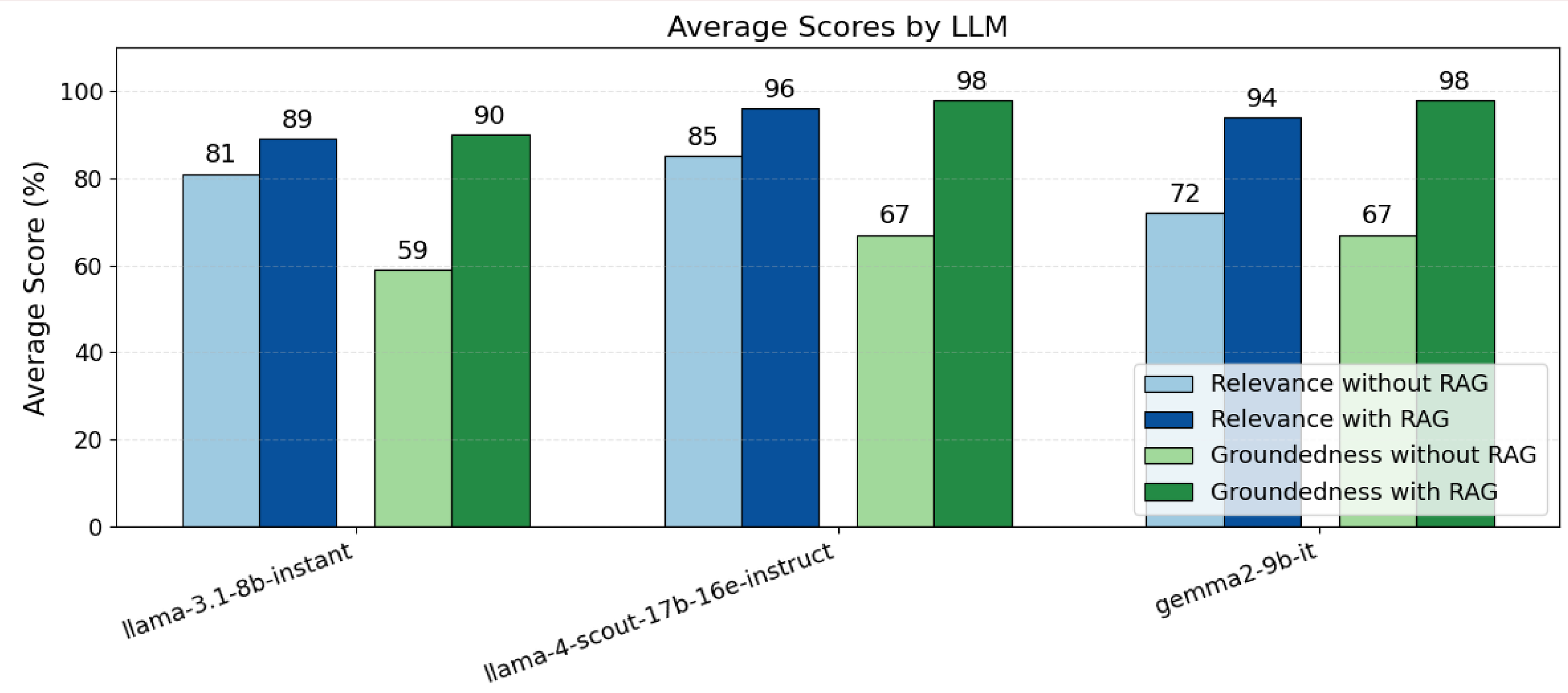
Six tolerance ranges, E1–E6, measure how closely generated FKGL matches the target level.

RAG Improves Quality of Generated Text



Across prompting strategies, RAG substantially improves both relevance and groundedness. The strongest gains are observed for groundedness, indicating that retrieved evidence helps reduce hallucinations and improves factual support.

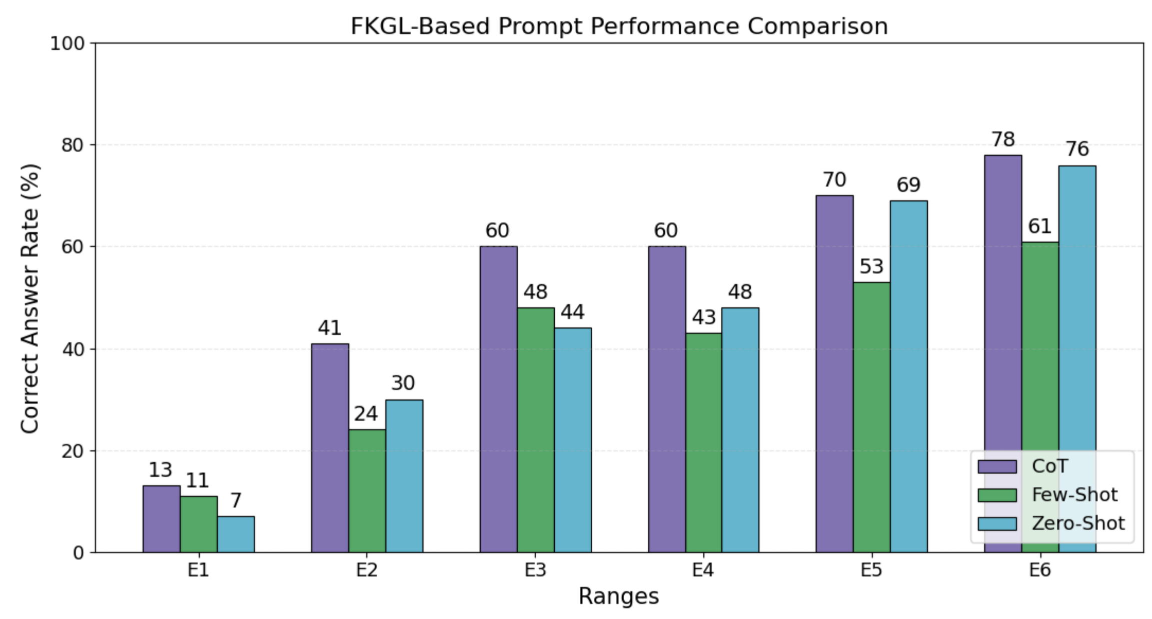
LLM Model Comparison

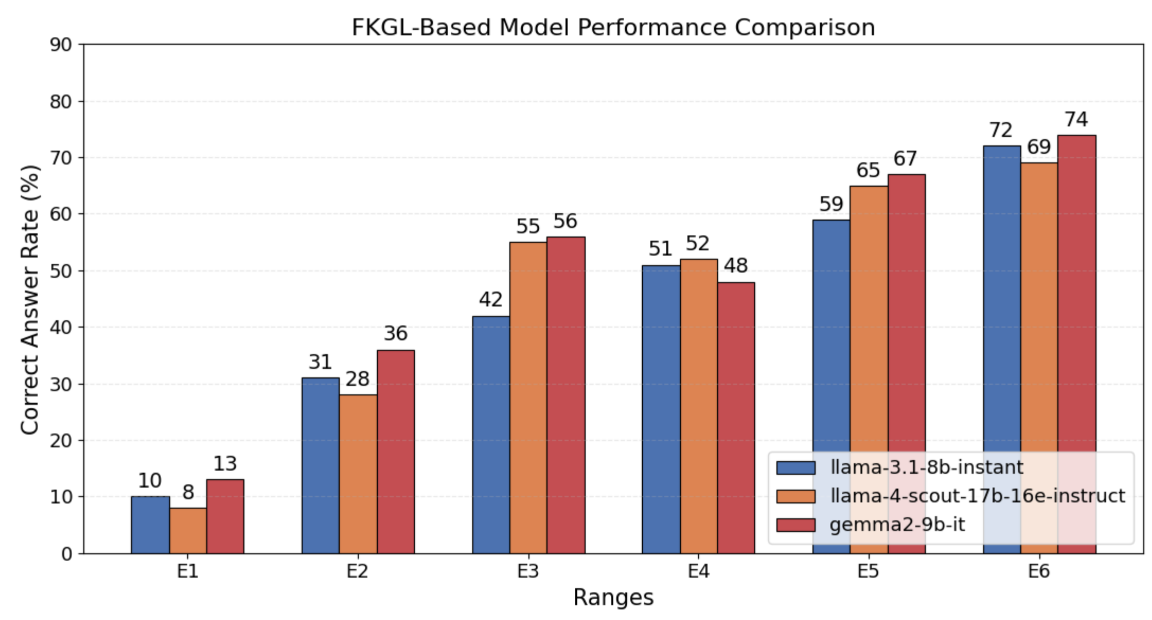


All tested models benefit from RAG. The improvement is systematic across LLaMA 4 Scout, LLaMA 3.1–8B–Instant, and Gemma2–9B. Retrieval reduces the need for the model to rely only on parametric knowledge.

Readability Control with FKGL

Range	Interpretation
E1	Exact match: the floored generated FKGL must equal the target FKGL.
E2	Close match: the generated FKGL may differ from the target by at most one grade level.
E3	Relaxed close match: the generated FKGL may differ by at most two grade levels.
E4	Broad band match: both target and generated FKGL must fall in the same band: low, middle, or high.
E5	Wider band match: similar to E4, but allows one additional neighboring grade level near each band boundary.
E6	Widest band match: allows broader deviations around the low, middle, and high readability bands.





Exact FKGL matching is difficult. Under the strictest E1 tolerance, the system matches the target readability level for only about **10%** of questions. Performance improves under relaxed tolerance ranges, with nearly half of generated answers matching the target within E3.

Main Findings

26–35 pp	**~10%**	**~50%**
groundedness gain with RAG	strict FKGL matching under E1	FKGL matching under E3

- **RAG** consistently increases relevance and groundedness.
- Groundedness improves the most, showing reduced hallucination risk.
- Prompting differences are clearer without RAG; with RAG, performance converges.
- Chain-of-Thought achieves the strongest overall FKGL results among prompting strategies.
- **Gemma2–9B** typically performs best in FKGL tolerance evaluation.

Acknowledgments

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